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Optical system

Abstract

An optical system includes an optical module with a main axis is provided. The optical module includes a fixed portion, a movable portion, a driving mechanism, and a supporting assembly. The movable portion is connected to an optical element and is movable relative to the fixed portion. The driving mechanism drives the movable portion to move relative to the fixed portion. The supporting assembly is connected to the movable portion and the fixed portion. When viewed along a direction that is parallel with the main axis, the fixing portion is a polygonal structure with a first side, a second side, a third side, and a fourth side. The first side is parallel with the third side, the second side is parallel with the fourth side, and the first side is not parallel with the second side.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation Application of U.S. patent application Ser. No. 17/155,964, filed on Jan. 22, 2021, which claims the benefit of U.S. Provisional Application No. 62/964,377, filed on Jan. 22, 2020, U.S. Provisional Application No. 63/017,313, filed on Apr. 29, 2020, U.S. Provisional Application No. 63/056,183, filed on Jul. 24, 2020, U.S. Provisional Application No. 63/058,932, filed on Jul. 30, 2020, and U.S. Provisional Application No. 63/121,415, filed on Dec. 4, 2020, the entirety of which are incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an optical system.

Description of the Related Art

(2) As technology has developed, optical elements and optical systems, which are used for driving optical elements, have become miniaturized. Many electronic devices (such as tablet computers and smartphones) are equipped with at least one optical element, at least one optical system, and at least one light-detection element for capturing images and recording videos. When a user uses an electronic device, shock or vibration may occur, and this may cause the images or videos to come out blurry. However, as the demand for higher quality in images and videos is increasing, an optical system that is able to perform displacement-correction and shake-compensation has been developed.

(3) The optical system may drive the optical element to move along a direction that is parallel with the optical axis to autofocus (AF) on the scene to be shot. Additionally, the optical system may also drive the optical element to move along a direction that is perpendicular to the optical axis to perform optical image stabilization (OIS), which compensates for the deviation of the image caused by shaking or impact, and solve the problem of blurry images and videos. AF and OIS may enhance the quality of the image.

BRIEF SUMMARY OF THE INVENTION

(4) Some embodiments of the invention provide an optical system including an optical module with a main axis. The optical module includes a fixed portion, a movable portion, a driving mechanism, and a supporting assembly. The movable portion is connected to an optical element and is movable relative to the fixed portion. The driving mechanism drives the movable portion to move relative to the fixed portion. The supporting assembly is connected to the movable portion and the fixed portion. When viewed along a direction that is parallel with the main axis, the fixed portion is a polygonal structure with a first side, a second side, a third side, and a fourth side. The first side is parallel with the third side, the second side is parallel with the fourth side, and the first side is not parallel with the second side.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:
- (2) FIG. 1 is a perspective view of the optical system.
- (3) FIG. 2 is a schematic diagram of optical elements in the optical system.
- (4) FIG. 3 is a perspective view of an optical module according to an embodiment of the disclosure.
- (5) FIG. 4 is an exploded view of the optical module according to an embodiment of the disclosure.
- (6) FIG. 5 is a perspective view of a partial structure of the optical module according to an embodiment of the disclosure.
- (7) FIG. 6 is a cross-sectional view of the optical module taken along the line A-A' in FIG. 3.
- (8) FIG. 7 is a schematic diagram of a partial structure of an optical module according to another embodiment of the disclosure.
- (9) FIG. 8 is an enlarged schematic diagram of the framed part of FIG. 5.
- (10) FIG. 9 is a schematic cross-sectional view of the optical module taken along the line B-B' in FIG. 3;
- (11) FIG. 10 is a perspective view of a partial structure of an optical module according to another embodiment of the disclosure.
- (12) FIG. 11 is a cross-sectional view of the optical module taken along the line A-A' in FIG. 10.
- (13) FIG. 12 is a block diagram of the circuit assembly of the optical module according to an embodiment of the disclosure.
- (14) FIG. 13 is a schematic diagram of a partial structure of an optical module according to an embodiment of the disclosure.
- (15) FIG. 14 is a schematic diagram of a partial structure of another optical module according to an embodiment of the disclosure.
- (16) FIG. 15 is a schematic diagram of a partial structure of another optical module according to an embodiment of the disclosure.
- (17) FIG. 16 is a schematic diagram of a partial structure of another optical module according to an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

- (18) The following description is made for the purpose of illustrating the general principles of the invention and should not be taken in a limiting sense. The scope of the invention is best determined by reference to the appended claims.
- (19) According to some embodiments, an optical system 1 is provided. FIG. 1 is a perspective view of the optical system 1. FIG. 2 is a schematic diagram of the optical elements in the optical system 1. For simplicity, only the optical elements are shown in FIG. 2. The optical system 1 may be a periscope optical system. The optical system 1 includes a first optical module 101, a second optical module 102, a third optical module 103, a fourth optical module 104, and a fifth optical module

105.

(20) A light above the first optical module **101** enters the first optical module **101** along a first incident direction L1, and is adjusted to pass through the third optical module **103**, the second optical module **102**, and the fourth optical module **104** sequentially along a second incident direction L2 by the first optical module **101**, and the optical path of the light is adjusted to be along a third incident direction L3 in the fifth optical module **105**, and imaging in the fifth optical module **105**.

(21) The first optical module **101** and the fifth optical module **105** may include a first optical element **111** and a fifth optical element **115**, respectively. The first optical element **111** and the fifth optical element **115** may be a prism, a mirror, a refractive prism, or a beam splitter, etc. By rotating the first optical element **111** and the fifth optical element **115**, the optical path of the light may be changed. The second optical module **102**, the third optical module **103**, and the fourth optical module **104** may respectively include a second optical element **112**, a third optical element **113**, and a fourth optical element **114**. The second optical module **102**, the third optical module **103**, and the fourth optical module **104** may drive the second optical element **112**, the third optical element **113**, and the fourth optical element **114** respectively. The second optical element **112**, the third optical element **113**, and the fourth optical element **114** may be one or more lenses, optical lenses, etc., and are made of materials such as glass and resin. The fifth optical module **105** may also include a sixth optical element **116**. The sixth optical element **116** may be an image sensor (or called a photosensitive element), etc., for example, a charge-coupled Device (CCD).

(22) In some embodiments, an optical element corresponding to a focal length of the first optical element **111** that is not zero (not shown, for example, one or more lens, optical lens, etc.) may be provided above the first optical element **111**. In other words, the optical element whose focal length is not zero may be fixedly connected to the first optical element **111**, and arranged along the first incident direction L1 with the first optical element **111**, and the shooting effect of the optical system **1** may be enhanced by increasing the quantity of the optical element.

(23) In some embodiments, the first optical module **101** and the fifth optical module **105** may perform yawing and pitching, respectively. In some embodiments, the first optical module **101** may also perform pitching, and the fifth optical module **105** may perform yawing. In some embodiments, the first optical module **101** and the fifth optical module **105** may both perform pitching. In some embodiments, the first optical module **101** and the fifth optical module **105** may both perform yawing. In some embodiments, the second optical module **102** and the third optical module **103** may achieve the functions of zooming and auto focusing (AF), respectively. In some embodiments, the second optical module **102** may also perform auto focusing, and the third optical module **103** may perform zooming. In other words, terms such as yawing, pitching, zooming, and auto focusing, etc., do not constitute limitations.

(24) In some embodiments, the fourth optical module **104** may achieve the function of Optical Image Stabilization (OIS). In some embodiments, the position of the fourth optical module **104** may be changed, for example, the fourth optical module **104** is disposed between the third optical module **103** and the fifth optical module **105**. In some embodiments, the fourth optical module **104** may be integrated into the second optical module **102** or the third optical module **103**, and the functions of auto focusing and optical image stabilization may be simultaneously achieved through a single second optical module **102** or a single third optical module **103**. In some embodiments, the fourth optical module **104** may be omitted.

(25) Refer to FIG. 3 to FIG. 7. FIG. 3 is a perspective view of an optical module **1000** according to an embodiment of the present disclosure. FIG. 4 is an exploded view of the optical module **1000** according to an embodiment of the disclosure. FIG. 5 is a perspective view a partial structure of the optical module **1000** according to an embodiment of the disclosure. FIG. 6 is a cross-sectional view of the optical module **1000** taken along the line A-A' in FIG. 3. FIG. 7 is a schematic diagram of a partial structure of an optical module **1000'** according to another embodiment of the disclosure. In

the following embodiments, the optical module **1000** has a similar structure to the fifth optical module **105** described above. The optical module **1000** has a main axis M, and includes an optical path adjustment element **1010**, an optical element **1020**, a fixed portion **1100**, a movable portion **1200**, a driving mechanism **1300**, a position sensing module **1400**, a supporting assembly **1500**, a circuit assembly **1600**, a dust-proof assembly **1700**, a heat dissipation assembly **1800**, and a damping assembly **1900** (shown in FIG. 9).

(26) The optical path adjustment element **1010** has a structure similar to the fifth optical element **115** mentioned above. In some embodiments, the optical adjustment element **1010** is a right-angle prism, but is not limited to this. The optical element **1020** has a structure similar to the sixth optical element **116** mentioned above. In some embodiments, the optical element **1020** is an image sensor, but is not limited to this.

(27) When viewed along a direction that is parallel with the main axis M, the fixed portion **1100** is a polygonal structure with a first side **1101**, a second side **1102**, a third side **1103** and a fourth side **1104**. The first side **1101** is parallel with the third side **1103**, the second side **1102** is parallel with the fourth side **1104**, and the first side **1101** and the second side **1102** are not parallel.

(28) As shown in FIG. 3 to FIG. 5, the fixed portion **1100** includes a base **1110**, an outer frame **1120**, and a frame **1130**. The base **1110** has a plate-like structure and is perpendicular to the main axis M. The outer frame **1120** and the base **1110** are arranged along the main axis M. The outer frame **1120** includes a top wall **1120T**, a first side wall **1121**, a second side wall **1122**, a third side wall **1123**, and a fourth side wall **1124**. The top wall **1120T** has a plate-like structure and is not parallel with the main axis M. In more detail, the top wall **1121T** is parallel with the base **1110**. The first side wall **1121** extends from an edge of the top wall **1120T** and is not parallel with the top wall **1120T**. When viewed along a direction that is parallel with the main axis M, the first side wall **1121** is disposed on the first side **1101**. The second side wall **1122** extends from an edge of the top wall **1120T** and is not parallel with the top wall **1120T**. When viewed along a direction that is parallel with the main axis M, the second side wall **1122** is disposed on the second side **1102**. The third side wall **1123** extends from an edge of the top wall **1120T** and is not parallel with the top wall **1120T**. When viewed along a direction that is parallel with the main axis M, the third side wall **1123** is disposed on the third side **1103**. The fourth side wall **1124** extends from an edge of the top wall **1120T** and is not parallel with the top wall **1120T**. The fourth side wall **1124** has an opening **1124O** corresponding to the incident light L. The opening **1124O** is disposed between the outer frame **1120** and the base **1110**. When viewed along a direction that is parallel with the main axis M, the fourth side wall **1124** is disposed on the fourth side **1104**.

(29) As shown in FIG. 6, the base **1110** and the outer frame **1120** form a first accommodating space S1. The first accommodating space S1 accommodates the movable portion **1200** and the frame **1130**. There is a second accommodating space S2 between the frame **1130** and the movable portion **1200** for accommodating the optical path adjustment element **1010**. Although in this embodiment, the frame **1130** is connected and fixed to the outer frame **1120**, and the optical path adjustment element **1010** is connected and fixed to the frame **1130**, it is not limited to this. In some embodiments, the frame **1130** may be included in the movable portion **1200**, and the frame **1130** and the optical path adjustment element **1010** may move relative to the fixed portion **1100**.

(30) As shown in FIG. 7, the optical module **1000'** has a similar structure to the optical module **1000**, except that the optical module **1000'** includes an anti-vibration driving mechanism **2000'** and a frame **1130'** has an inclined surface S' that may move relative to the fixed portion **1100'**. The optical path adjustment element **1010'** is fixedly disposed on the inclined surface S' of the frame **1130'**. When viewed along a direction that is parallel with the main axis M', the anti-vibration driving mechanism **2000'** is disposed on the second side **1102'**, and the anti-vibration driving mechanism **2000'** and the driving mechanism **1300'** at least partially overlap. The anti-vibration driving mechanism **2000'** may drive the optical path adjustment element **1010'** to move relative to the fixed portion **1100'**. In more detail, according to the configuration of the anti-vibration driving

mechanism **2000'**, the optical path adjustment element **1010'** may be controlled to be yawing and pitching. In the embodiment shown in FIG. 7, the anti-vibration driving mechanism **2000'** drives the optical path adjustment element **1010'** to rotate in a first rotating axis A' relative to the fixed portion **1100'**, and the first rotating axis A' is parallel with the second side **1102'**.

(31) Back to FIG. 3 to FIG. 6, the movable portion **1200** is connected to the optical element **1020** and may move relative to the fixed portion **1100**. The movable portion **1200** includes a pedestal **1210** and a holder **1220**. The pedestal **1210** is connected to the optical element **1020** and has a plate-like structure. The holder **1220** is fixedly disposed on the pedestal **1210**, and has a shielding portion **1221**, which is arranged near the fourth side **1104**. In more detail, when viewed along a direction that is parallel with the main axis M, the light enters the optical module **1000** from the fourth side **1104**, and enters the optical element **1020** through the optical path adjustment element **1010**. The optical path adjustment element **1010** is configured to adjust the light traveling in the direction that is parallel with the first side **1101** to travel in a direction that is parallel with the main axis M. The shielding portion **1221** is a protruding structure which protrudes along a direction that is parallel with the main axis M. When viewed along the direction that is parallel with the first side **1101**, the shielding portion **1221** and the optical element **1020** at least partially overlap, so a stray light may be shielded, and the stray light may not enter the optical element **1020**.

(32) The shielding portion **1221** may also be used as a stopping portion **1221**. An inner side wall located on the fourth side **1104** and close to the stopping portion **1221** may be used as a stopping surface **1124A**, so the stopping portion **1221** and the stopping surface **1124A** may form a stopping assembly **1600** configured to restrict the movable portion **1200** to move within a moving range relative to the fixed portion **1100**. In more detail, when the stopping portion **1221** touches the stopping surface **1124A**, the movable portion **1200** stops moving, so the movable portion **1200** moving along a direction that is not parallel with the main axis M may be restricted within a moving range relative to the fixed portion **1100**.

(33) The dust-proof assembly **1700** may be adhesive, glue or the like, which has a higher viscosity. The dust-proof assembly **1700** is at least partly disposed on the pedestal **1210** and at least partly disposed on the holder **1220**. When viewed along a direction that is parallel with the main axis M, the dust-proof assembly **1700** does not overlap the optical element **1020**, and the dust-proof assembly **1700** is located around the optical element **1020**. In more detail, the dust-proof assembly **1700** has a closed structure surrounding the optical element **1020**. Due to the higher viscosity of the dust-proof assembly **1700**, the dust-proof assembly **1700** may capture foreign objects and restrict the path of the foreign objects to prevent the foreign objects from contacting the optical element **1020**.

(34) The heat dissipation assembly **1800** is configured to improve the heat dissipation efficiency of the optical element **1020** and the driving mechanism **1300**. The heat dissipation assembly **1800** includes a first heat dissipation element **1810** and a second heat dissipation element **1920**. The first heat dissipation element **1810** has a plate-shaped structure with a thermal conductivity greater than $10 \text{ W}/(\text{m}\times\text{K})$, and may be made of carbon fiber, ceramic or metal. The first heat dissipation element **1810** is closer to the optical element **1020** than the top wall **1120T**, and the first heat dissipation element **1810** and the base **1110** may have an integrated structure.

(35) The second heat dissipation element **1820** has a plate-shaped structure with a thermal conductivity greater than $10 \text{ W}/(\text{m}\times\text{K})$, and is fixedly disposed on the pedestal **1210** and is at least partially embedded in the pedestal **1210**. The second heat dissipation element **1820** has a magnetically permeable material to improve the driving efficiency of the driving mechanism **1300**. In more detail, the second heat dissipation element **1820** corresponds to the driving mechanism **1300**, and because of the magnetic material, the second heat dissipation element **1820** may generate an attractive force or a repulsive force to improve the driving efficiency of the driving mechanism **1300**. The second heat dissipation element **1820** is parallel with the first heat dissipation element **1810**, and there is a gap greater than zero between the first heat dissipation element **1810** and the

second heat dissipation element **1820**.

(36) The driving mechanism **1300** drives the movable portion **1200** to move relative to the fixed portion **1100**. The driving mechanism **1300** includes a first driving assembly **1310**, a second driving assembly **1320**, and a third driving assembly **1330**. The first driving assembly **1310** is disposed on the first side **1101**, and includes a first coil **1311**, a first magnetic element **1312**, a second coil **1313**, and a second magnetic element **1314**. The first magnetic element **1312** corresponds to the first coil **1311**, and the first coil **1311** and the first magnetic element **1312** are arranged along a direction that is parallel with the main axis M. The second coil **1313** and the first coil **1311** are arranged along a first direction D1, and the first direction D1 is parallel with the first side **1101**. The second magnetic element **1314** corresponds to the second coil **1313**, and the second magnetic element **1314** and the first magnetic element **1312** are arranged along the first direction D1. The second driving assembly **1320** is disposed on the second side **1102**, and includes a third coil **1321**, a third magnetic element **1322**, a fourth coil **1323**, and a fourth magnetic element **1324**. The third magnetic element **1322** corresponds to the third coil **1321**, and the third coil **1321** and the third magnetic element **1322** are arranged along a direction that is parallel with the main axis M. The fourth coil **1323** and the third coil **1321** are arranged along the second direction D2, and the second direction D2 is parallel with the second side **1102**. The fourth magnetic element **1324** corresponds to the fourth coil **1323**. The third driving assembly **1330** is disposed on the third side **1103**, and includes a fifth coil **1331**, a fifth magnetic element **1332**, a sixth coil **1333**, and a sixth magnetic element **1334**. The fifth magnetic element **1332** corresponds to the fifth coil **1331**, and the fifth coil **1331** and the fifth magnetic element **1332** are arranged along a direction that is parallel with the main axis M. The sixth coil **1333** and the fifth coil **1331** is arranged along a third direction D3, and the third direction D3 is parallel with the third side **1103**. The sixth magnetic element **1334** corresponds to the sixth coil **1333**. A winding axis of the second coil **1313** is parallel with a winding axis of the first coil **1311**. The winding axis of the first coil **1311** is parallel with the main axis M. A winding axis of the fourth coil **1323** is parallel with a winding axis of the third coil **1321**. The winding axis of the fourth coil **1323** is parallel with the main axis M. A winding axis of the sixth coil **1333** is parallel with a winding axis of the fifth coils **1331**. The winding axis of the sixth coil **1333** is parallel with the main axis M. When viewed along a direction that is parallel with the main axis M, the driving mechanism **1300** is not disposed on the fourth side **1104**. The driving assembly is at least partially fixedly disposed on the holder **1220**.

(37) In some embodiments, the first coil **1311**, the second coil **1313**, the third coil **1321**, the fourth coil **1323**, the fifth coil **1331**, and the sixth coil **1333** are disposed on the holder **1220** of the movable portion **1200**, the first magnetic element **1312**, the second magnetic element **1314**, the third magnetic element **1322**, the fourth magnetic element **1324**, the fifth magnetic element **1332**, and the sixth magnetic element **1334** are disposed on the frame **1130** of the fixed portion **1100**, but not limited to this. In some embodiments, the first coil **1311**, the second coil **1313**, the third coil **1321**, the fourth coil **1323**, the fifth coil **1331**, and the sixth coil **1333** are disposed on the frame **1130** of the fixed portion **1100**, and the first magnetic element **1312**, the second magnetic element **1314**, the third magnetic element **1322**, the fourth magnetic element **1324**, the fifth magnetic element **1332**, and the sixth magnetic element **1334** are disposed on the holder **1220** of the movable portion **1200**.

(38) In addition, in the embodiment of FIG. 4, the first driving assembly **1310**, the second driving assembly **1320**, and the third driving assembly **1330** respectively include two magnetic elements, but it is not limited to this. In some embodiments, the first magnetic element **1312** and the second magnetic element **1314** have an integrally formed structure, the third magnetic element **1322** and the fourth magnetic element **1324** have an integrally formed structure, and the fifth magnetic element **1332** and the sixth magnetic element **1334** have an integrated structure.

(39) The position sensing module **1400** is configured to sense the movement of the movable portion **1200** relative to the fixed portion **1100**, and includes a first position sensing assembly **1410**,

a second position sensing assembly **1420**, and a third position sensing assembly **1430**. The first position sensing assembly **1410** has a first reference element **1411** and a first sensing element **1412**, and the first sensing element **1412** is configured to sense a first magnetic field generated by the first reference element **1411**. When viewed along a direction that is parallel with the main axis M, the first sensing element **1412** and the first reference element **1411** are disposed on the first side **1101**. The second position sensing assembly **1420** has a second reference element **1421** and a second sensing element **1422**, and the second sensing element **1422** is configured to sense the second magnetic field generated by the second reference element **1421**. When viewed along a direction that is parallel with the main axis M, the second sensing element **1422** and the second reference element **1421** are disposed on the second side **1102**. The third position sensing assembly **1430** has a third reference element **1431** and a third sensing element **1432**. The third sensing element **1432** is configured to sense a third magnetic field generated by the third reference element **1431**. When viewed in a direction that is parallel with the main axis M, the third sensing element **1432** and the third reference element **1431** are disposed on the third side **1103**. When viewed along a direction that is parallel with the main axis M, the position sensing module **1400** is not disposed on the fourth side **1104**.

(40) In some embodiments, the first reference element **1411**, the second reference element **1421**, and the third reference element **1431** are respectively magnetic elements, which are disposed on the frame **1130**. When viewed along a direction that is perpendicular to the main axis M, the first reference element **1411** is disposed between the first magnetic element **1312** and the second magnetic element **1314**, the second reference element **1421** is disposed between the third magnetic elements **1322** and the fourth magnetic element **1324**, the third reference element **1431** is arranged between the fifth magnetic element **1332** and the sixth magnetic element **1334**. The first sensing element **1412**, the second sensing element **1422**, and the third sensing element **1432** may be, for example, a Hall effect sensor, a magnetoresistive (MR) sensor, or a Fluxgate, etc. The first sensing element **1412**, the second sensing element **1422**, and the third sensing element **1432** are disposed on the holder **1220**. When viewed along a direction that is perpendicular to the main axis M, the first sensing element **1412** is disposed between the first coil **1311** and the second coil **1313**, the second sensing element **1422** is located between the third coil **1321** and the fourth coil **1323**, and the third sensing element **1432** is located between the fifth coil **1331** and the sixth coil **1333** to respectively sense the first magnetic field of the first reference element **1411**, the second magnetic field of the second reference element **1421**, and the third magnetic field of the third reference element **1431** to obtain the position of the holder **1220** relative to the frame **1130**.

(41) In some embodiments, three sets of position sensing assemblies are provided (a first position sensing assembly **1410**, a second position sensing assembly **1420**, and a third position sensing assembly **1430**) to sense the movement and rotation of the movable portion **1200** relative to the fixed portion **1100**, but not limited to this. In some embodiments, only two sets of position sensing assemblies that are not parallel with each other may be provided to sense the movement of the movable portion **1200** relative to the fixed portion **1100**.

(42) The above-mentioned example with the position sensing module **1400** is driven by a closed-loop control. That is, the position sensing module **1400** is used to sense whether the movable portion **1200** reaches the expected position, and if it does not reach the expected position, a command may be made by a controller of the driving mechanism **1300** to make corrections until the movable portion **1200** reaches the expected position. However, the present disclosure is not limited to this. It may also be driven by an open-loop control without the position sensing module **1400** with the position feedback, and by establishing a database in advance, then the movable portion **1200** may be driven directly to the expected position.

(43) Next, refer to FIG. 4 to FIG. 6, and FIG. 8. FIG. 8 is an enlarged schematic view of the framed part of FIG. 5. The supporting assembly **1500** is connected to the movable portion **1200** and the fixed portion **1100**. The supporting assembly **1500** includes a first elastic element **1510** and a

second elastic element **1520**. The first elastic element **1510** has a long strip structure extending along a direction that is parallel with the main axis M, and pass through the frame **1130** and the holder **1220**. The first elastic element **1510** has a first end portion **1511** and a second end portion **1512**. The first end portion **1511** is fixed to the frame **1130**, and is located between the frame **1130** and the top wall **1120T**, but is not in direct contact with the top wall **1120T**. In more detail, the frame **1130** has a groove **1131** to accommodate the first end portion **1511**. The second end portion **1512** is connected to the second elastic element **1520**.

(44) As shown in FIG. 8, the second elastic element **1520** has a plate-like structure, which is not parallel with the first elastic element **1510**. The second elastic element **1520** has an impact absorbing portion **1521** and a fixed end portion **1522**. The impact absorbing portion **1521** is connected to the second end portion **1512**. The impact absorbing portion **1521** absorbs the impact of the first elastic element **1510**. The fixed end portion **1522** is fixed to the holder **1220**. Therefore, the second elastic element **1520** contacts to the holder **1220**, and is not in direct contact with the pedestal **1210**. When viewed along a direction that is perpendicular to the main axis M, the second elastic element **1520** is located between the pedestal **1210** and the holder **1220**. When viewed along a direction that is parallel with the main axis M, the first elastic element **1510**, the pedestal **1210** and the holder **1220** at least partially overlap.

(45) Refer to FIG. 4 and FIG. 9. FIG. 9 is a schematic cross-sectional view of the optical module **1000** along the line B-B' in FIG. 3. The damping assembly **1900** may have a gel material, and includes a first damping element **1910**, a second damping element **1920**, a third damping element **1930**, a fourth damping element **1940**, and a fifth damping element **1950**. The first damping element **1910** is in direct contact with the first elastic element **1510** and the movable portion **1200**. In more detail, the first damping element **1910** is disposed between the second end portion **1512** and the pedestal **1210**. The second damping element **1920** is in direct contact with the first elastic element **1510** and the fixing portion **1100**. In more detail, the second damping element **1920** is disposed between the first end portion **1511** and the top wall **1120T** to avoid short-circuit causing by the first end portion **1511** contacts the top wall **1120T**.

(46) The third damping element **1930** is in direct contact with the first elastic element **1510** and the frame **1130**. The fourth damping element **1940** is in direct contact with the first elastic element **1510** and the movable portion **1200**. In more detail, the fourth damping element **1940** is disposed between the first elastic element **1510** and the holder **1210**. The fifth damping element **1950** is in direct contact with the movable portion **1200** and the fixed portion **1100**. In more detail, the fifth damping element **1950** is disposed between the movable portion **1200** and the base **1110**. By providing the damping assembly **1900**, the connection between the supporting assembly **1500** and the movable portion **1200** and the fixed portion **1100** may be strengthened, and the movement of the movable portion **1200** relative to the fixed portion **1100** may be stabilized.

(47) In addition, an intermediate element **1530''** may be used to replace the first elastic element **1510** and the second elastic element **1520** to form the supporting assembly **1500''**. As shown in FIG. 10 and FIG. 11, FIG. 10 is a perspective view of a partial structure of an optical module **1000''** according to another embodiment of the disclosure, and FIG. 11 is a cross-sectional view of the optical module **1000''** taken along the line A-A' in FIG. 10. The supporting assembly **1500''** includes eight intermediate elements **1530''**, and each intermediate element **1530''** has a spherical structure. In this embodiment, the four corners of the upper and lower sides of the holder **1220''** are provided with holder recesses, and the frame **1130''** is provided with four frame recesses corresponding to the holder recesses on the upper side of the holder **1220''**, and the base **1110''** is provided with four base recesses corresponding to the holder recesses on the lower side of the holder **1220''**. The holder recesses, frame recesses, and base recesses respectively accommodate part of the eight intermediate elements **1530''**. When the driving mechanism **1300''** drives the movable portion **1200''** to move, the movable portion **1200''** may move relative to the fixed portion **1100''** by the intermediate element **1530''**.

(48) Refer to FIG. 4 and FIG. 12. FIG. 12 is a block diagram of the circuit assembly **1600** in the optical module **1000** according to an embodiment of the disclosure. The circuit assembly **1600** is electrically connected to a circuit E located outside the optical module **1000**. The circuit assembly **1600** includes a first circuit **1610**, a second circuit **1620**, and a third circuit **1630**, and an external circuit **1640**. The first circuit **1610** is fixedly disposed on the pedestal **1210**, and is integrally formed with the pedestal **1210**. The second circuit **1620** is fixedly disposed on the holder **1220**, and is at least partially embedded in the holder **1220**. The third circuit **1630** is fixedly disposed on the frame **1130** and is at least partially embedded in the frame **1130**. The third circuit **1630** has a first connecting portion **1631**, which is made of metal and is at least partially embedded in frame **1130**, and may connect to the first end portion **1511** of the first elastic element **1510**. The external circuit **1640** is connected to the circuit E outside the optical module **1000** and is connected to the first circuit **1610**. The external circuit **1640** is flexible, for example, it can be a flexible printed circuit board (FPC). The external circuit **1640** has a plurality of holes H, which makes the whole body of the external circuit **1640** easier to be bent and not easier to be broken, and also makes the movable portion **1200** easier to move.

(49) In addition, in the embodiment of FIG. 10, at least one of the first circuit **1610''**, the second circuit **1620''**, and the third circuit **1630''** (not shown for embedding in the frame **1130'**) has a magnetically permeable material. In more detail, the second circuit **1620''** has a magnetically permeable material and corresponds to the driving mechanism **1300''**, so that the movable portion **1200''** may move close to the intermediate element **1530''** by the attraction force, thus moving more stably relative to the fixed portion **1100''**.

(50) Next, refer to FIG. 13 to FIG. 16, several different electrical connections will be described. FIG. 13 is a schematic diagram of a partial structure of an optical module **1000** according to an embodiment of the disclosure. FIG. 14 to FIG. 16 are schematic diagrams of a partial structure of an optical module **1000A** according to an embodiment of the disclosure.

(51) In the embodiment of FIG. 13, the coils are disposed on the holder **1220**, and the electrical connection is illustrated by taking the first driving assembly **1310** as an example, the second driving assembly **1320** and the third driving assembly **1330** may be electrically connected in the same way. The first driving assembly **1310** is electrically connected to the second circuit **1620**, and is electrically connected to the first circuit **1610** via the second circuit **1620**. In more detail, the first coil **1311** and the second coil **1313** are connected to the contacts of the second circuit **1620** exposed on the surface of the holder **1220** by welding, etc., and the partial contacts of the second circuit **1620** and the partial contacts of the first circuit **1610** are arranged in cooperation. Therefore, when the holder **1220** is assembled to the pedestal **1210**, the second circuit **1620** and the first circuit **1610** are electrically connected at the same time. Thus, the first driving assembly **1310** is electrically connected to the external circuit **1640** via the second circuit **1620** and the first circuit **1610** in sequence, and then electrically connected to the circuit E outside the optical module **1000** via the external circuit **1640**. The second driving assembly **1320** and the third driving assembly **1330** are also electrically connected in a similar way.

(52) In the embodiment shown in FIG. 14, the coils of the optical module **1000A** are disposed on the frame **1130A**, and the electrical connection is illustrated by taking the second driving assembly **1320A** as an example. The first driving assembly **1310A** and the third driving assembly **1330A** may be electrically connected in the same way. The third coil **1321A** and the fourth coil **1323A** of the second driving assembly **1320A** are welded to the contacts (not shown) of the third circuit **1630A** on the frame **1130A**, and are electrically connected to the third circuit **1630A**. The first connecting portion **1631A** of the third circuit **1630A** is electrically connected to the first end portion **1511A** of the first elastic element **1510A**, and the first elastic element **1510A** is electrically connected to the second elastic element. The second elastic element is connected to the contacts (not shown) of the second circuit **1620A** disposed on the holder **1220A**. Then, as in the embodiment of FIG. 13, the partial contacts of the second circuit **1620** and the partial contacts of the first circuit **1610** are

arranged in cooperation, so that the second circuit **1620A** is electrically connected to the first circuit **1610A**, and electrically connected the circuit E outside the optical module **1000A** via the external circuit **1640A**. Therefore, the second driving assembly **1320A** is electrically connected to the circuit E outside the optical module **1000A** passes via the third circuit **1630A**, the first elastic element **1510A**, the second elastic element A, the second circuit **1620A**, the first circuit **1610A**, and the external circuit **1640A** in sequence.

(53) In the embodiment of FIG. **15**, an intermediate circuit **1650A** may also be provided on the frame **1130A**, and the intermediate circuit **1650A** may be a flexible printed circuit board with flexibility. The second sensing element **1422A** may be electrically connected to the second circuit **1620A** via the intermediate circuit **1650A**, and then electrically connected to the circuit E outside the optical module **1000A** via the first circuit **1610A** and the external circuit **1640A** in sequence.

(54) In the embodiment of FIG. **16**, the second driving assembly **1320A** may also be directly connected to the external circuit **1640A** via the first elastic element **1510A** to be electrically connected to the circuit E outside the optical module **1000A**. In other words, another intermediate circuit **1650A'** may be provided on the top of the frame **1130A**. The intermediate circuit **1650A'** may be a flexible printed circuit board with flexibility, which is directly connected to the first elastic element **1510A** and the external circuit **1640A**. Therefore, the second driving assembly **1320A** is connected to the circuit E outside the optical module **1000A** via the third circuit **1630A**, the first elastic element **1510A**, the intermediate circuit **1650A'**, and the external circuit **1640A** in sequence.

(55) However, the electrical connection is not limited to the above-mentioned embodiments, and the configuration of the optical module **1000** may be changed in combination with the electrical connection described above as required, so that the three driving assemblies may be electrically connected in different ways. For example, in an optical module **1000B** (not shown), the first driving assembly **1310B** is connected to the circuit E outside the optical module **1000B** via the second circuit **1620B**, the first circuit **1610B**, and the external circuit **1640B** in sequence. The second driving assembly **1320B** is connected to the circuit E outside the optical module **1000B** via the third circuit **1630B**, the first elastic element **1510B**, the second elastic element B, the second circuit **1620B**, the first circuit **1610B**, and the external circuit **1640B** in sequence. The third driving assembly **1330B** is connected to the circuit E outside the optical module **1000B** via the third circuit **1630B**, the first elastic element **1510B**, the intermediate circuit **1650B**, and the external circuit **1640B** in sequence.

(56) In addition, in the embodiment in which the intermediate element constitutes the supporting assembly, if the coils are disposed on the holder **1220''**, as shown in FIG. **10**, the electrical connection is similar to the embodiment in FIG. **13**. The first driving assembly **1310''** is electrically connected to the circuit E outside the optical module **1000''** via the second circuit **1620''** and the first circuit **1610''** and the external circuit **1640''** in sequence.

(57) If the coils is disposed on the frame **1130''** (not shown), the electrical connection is similar to the embodiment in FIG. **15** and/or FIG. **16**. The first driving assembly **1310''** is electrically connected the intermediate circuit **1650''** via the third circuit **1630''**, and the intermediate circuit **1650''** may be electrically connected to one of the first circuit **1610''** and the external circuit **1640''**, and then, electrically connected the circuit E outside the optical module **1000''** via the external circuit **1640''**. In this embodiment, the first driving assembly **1310''** is electrically connected to the circuit E outside the optical module **1000''** via the third circuit **1630''**, the intermediate circuit **1650''** and the external circuit **1640''** in sequence.

(58) As described above, an embodiment of the present invention provides an optical system includes an optical module with a main axis. The optical module includes a fixed portion, a movable portion, a driving mechanism, and a supporting assembly. The movable portion is connected to an optical element and is movable relative to the fixed portion. The driving mechanism drives the movable portion to move relative to the fixed portion. The supporting

assembly is connected to the movable portion and the fixed portion. When viewed along a direction that is parallel with the main axis, the fixed portion is a polygonal structure with a first side, a second side, a third side, and a fourth side, the first side is parallel with the third side, the second side is parallel with the fourth side, the first side is not parallel with the second side. The special position and size relationship of each element disclosed in the present invention may enable the optical element driving mechanism to achieve a specific direction of thinning and overall miniaturization. In addition, by applying with different optical modules, the optical element driving mechanism may further improve the optical quality (such as shooting quality or depth sensing accuracy, etc.).

(59) While the invention has been described by way of example and in terms of the preferred embodiments, it should be understood that the invention is not limited to the disclosed embodiments. On the contrary, it is intended to cover various modifications and similar arrangements (as would be apparent to those skilled in the art). Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

Claims

1. An optical system, comprising: an optical module with a main axis, comprising: a fixed portion; a movable portion connected to an optical element and movable relative to the fixed portion; a driving mechanism driving the movable portion to move relative to the fixed portion; a supporting assembly connected to the movable portion and the fixed portion; and a circuit assembly electrically connected to a circuit located outside the optical module; wherein when viewed along a direction that is parallel with the main axis, the fixed portion is a polygonal structure with a first side, a second side, a third side, and a fourth side; wherein the first side is parallel with the third side, the second side is parallel with the fourth side, and the first side is not parallel with the second side; wherein the fixed portion comprises an outer frame and a frame connected to the outer frame, a first accommodating space is formed to accommodate the movable portion and the frame, and the outer frame comprises: a top wall having a plate-like structure and not parallel with the main axis; a first side wall extending from a first edge of the top wall and not parallel with the top wall; a second side wall extending from a second edge of the top wall and not parallel with the top wall; a third side wall extending from a third edge of the top wall and not parallel with the top wall; and a fourth side wall extending from a fourth edge of the top wall and not parallel with the top wall, wherein the fourth side wall has an opening corresponding to a light; wherein the movable portion comprises: a pedestal connected to the optical element and having a plate-like structure; and a holder fixedly disposed on the pedestal; wherein the circuit assembly comprises: a first circuit fixedly disposed on the pedestal and integrally formed with the pedestal; a second circuit fixedly disposed on the holder and at least partially embedded in the holder; a third circuit fixedly disposed on the frame and at least partially buried in the frame; and an external circuit connected to the circuit outside the optical module.

2. The optical system as claimed in claim 1, wherein the fixed portion further comprises a base having a plate-like structure and perpendicular to the main axis, and the base and the outer frame are arranged along the main axis; wherein the top wall is parallel with the base; wherein when viewed along a direction that is parallel with the main axis, the first side wall is disposed on the first side; wherein when viewed along a direction that is parallel with the main axis, the second side wall is disposed on the second side; wherein when viewed along a direction that is parallel with the main axis, the third side wall is disposed at the third side; wherein the opening is formed between the outer frame and the base, when viewed along a direction that is parallel with the main axis, the fourth side wall is disposed on the fourth side.

3. The optical system as claimed in claim 1, wherein the optical module further comprises an

optical path adjustment element connected to the frame, and a second accommodating space is formed between the frame and the movable portion to accommodate the optical path adjustment element.

4. The optical system as claimed in claim 1, wherein the driving mechanism comprises a first driving assembly disposed on the first side; wherein the first driving assembly is electrically connected to the second circuit; wherein the first driving assembly is electrically connected to the first circuit via the second circuit; wherein the first driving assembly is electrically connected to the external circuit via the second circuit and the first circuit in sequence; wherein the first driving assembly is electrically connected to the circuit outside the optical module via the second circuit, the first circuit, and the external circuit in sequence.

5. The optical system as claimed in claim 1, wherein the external circuit is flexible, and the external circuit has a plurality of holes.

6. The optical system as claimed in claim 1, wherein at least one of the first circuit, the second circuit, and the third circuit has a magnetically permeable material, and the second circuit has a magnetically permeable material and corresponds to the driving mechanism.

7. The optical system as claimed in claim 2, wherein the supporting assembly comprises a plurality of intermediate elements each having a spherical structure; wherein an upper side and a lower side of the holder are provided with a plurality of holder recesses, the frame is provided with a plurality of frame recesses corresponding to the holder recesses on the upper side of the holder, and the base is provided with a plurality of base recesses corresponding to the holder recesses on the lower side of the holder; wherein the holder recesses, the frame recesses, and the base recesses respectively accommodate part of the intermediate elements.

8. The optical system as claimed in claim 1, wherein the driving mechanism comprises a first driving assembly disposed on the first side, and the circuit assembly further comprises an intermediate circuit connected to one of the first circuit and the external circuit; wherein the first driving assembly is electrically connected to the third circuit; wherein the first driving assembly is electrically connected to the circuit outside the optical module via the intermediate circuit; wherein the first driving assembly is electrically connected to the intermediate circuit via the third circuit; wherein the first driving assembly is electrically connected to the circuit outside the optical module via the second circuit; wherein the first driving assembly is electrically connected to one of the first circuit and the external circuit via the intermediate circuit; wherein the first driving assembly is electrically connected to the circuit outside the optical module via the external circuit; wherein the first driving assembly is electrically connected to the circuit outside the optical module via the third circuit, the intermediate circuit and the external circuit in sequence.

9. The optical system as claimed in claim 2, wherein the supporting assembly comprises: a first elastic element having a long strip structure and extending along a direction that is parallel with the main axis; and a second elastic element having a plate-like structure that is not parallel with the first elastic element, in contact with the holder and not in direct contact with the base, and having an impact absorbing portion that absorbs the impact of the first elastic element; wherein when viewed along a direction that is perpendicular to the main axis, the second elastic element is located between the pedestal and the holder; wherein when viewed along a direction that is parallel with the main axis, the first elastic element, the pedestal, and the holder at least partially overlap.

10. The optical system as claimed in claim 9, wherein the optical module further comprises a damping assembly configured to stabilize the movement of the movable portion relative to the fixed portion, comprising: a first damping element in direct contact with the first elastic element and the movable portion; a second damping element in direct contact with the first elastic element and the fixing portion; a third damping element in direct contact with the first elastic element and the frame; a fourth damping element in direct contact with the first elastic element and the movable portion; and a fifth damping element in direct contact with the movable portion and the fixed portion, wherein the fifth damping element is in direct contact with the base, and the fifth damping

element is located between the movable portion and the base.

11. The optical system as claimed in claim 9, wherein the driving mechanism further comprises a second driving assembly disposed on the second side; wherein the second driving assembly is electrically connected to the third circuit; wherein the second driving assembly is electrically connected to the circuit outside the optical module via the first elastic element; wherein the second driving assembly is electrically connected to the first elastic element via the third circuit; wherein the second driving assembly is electrically connected to the circuit outside the optical module via the second elastic element; wherein the second driving assembly is electrically connected to the second elastic element via the first elastic element; wherein the second driving assembly is electrically connected to the circuit outside the optical module via the second circuit; wherein the second driving assembly is electrically connected to the second circuit via at least one of the first elastic element and the second elastic element; wherein the second driving assembly is electrically connected to the first circuit via the second circuit; wherein the second driving assembly is electrically connected to the external circuit via the first circuit; wherein the second driving assembly is electrically connected to the circuit outside the optical module via the external circuit; wherein the second driving assembly is electrically connected to the circuit outside the optical module via the third circuit, the first elastic element, the second circuit, the first circuit, and the external circuit in sequence.

12. The optical system as claimed in claim 9, wherein the frame has a groove, the first elastic element has a first end portion fixed to the frame, the first end portion is received in the groove and located between the frame and the top wall, and the first end portion is not in direct contact with the top wall.

13. The optical system as claimed in claim 12, wherein the third circuit has a first connecting portion connected to the first end portion, the first connecting portion is made of metal material, and the first connecting portion is at least partially embedded in the frame.

14. The optical system as claimed in claim 2, wherein the optical module further comprises a dust-proof assembly to limit path of foreign objects to prevent foreign objects from contacting the optical element; wherein the dust-proof assembly has a higher viscosity to catch foreign objects; wherein when viewed along a direction that is parallel with the main axis, the dust-proof assembly and the optical element do not overlap, the dust-proof assembly is located around the optical element, and the dust-proof assembly has a closed structure surrounding the optical element; wherein the dust-proof assembly is at least partially disposed on the pedestal; wherein the dust-proof assembly is at least partially disposed on the holder.

15. The optical system as claimed in claim 2, wherein the optical module further comprises a heat dissipation assembly configured to improve the heat dissipation efficiency of the optical element and the driving mechanism, comprising: a first heat dissipation element having a plate-shaped structure, having a material of carbon fiber, ceramic or metal material, and having a thermal conductivity greater than $10 \text{ W}/(\text{m}\times\text{K})$, wherein the first heat dissipation element and the base have an integral structure, and the first heat dissipation element is closer to the optical element than the top wall; and a second heat dissipation element having a plate-shaped structure, having a magnetically permeable material to improve the driving efficiency of the driving mechanism, and having a thermal conductivity greater than $10 \text{ W}/(\text{m}\times\text{K})$, wherein the second heat dissipation element is fixedly disposed on the pedestal and at least partially embedded in the pedestal, and the second heat dissipation element corresponds to the driving mechanism to generate an attractive force or a repulsive force; wherein the second heat dissipation element is parallel with the first heat dissipation element; wherein a gap greater than zero is formed between the first heat dissipation element and the second heat dissipation element.

16. The optical system as claimed in claim 1, wherein the optical module further comprises an optical path adjustment element configured to adjust a travel direction of the light entering the optical module from a direction that is parallel with the first side to a direction that is parallel with

the main axis, and wherein when viewed along a direction that is parallel with the main axis, the light enters the optical module from the fourth side.

17. The optical system as claimed in claim 16, wherein the optical module further comprises an anti-vibration driving mechanism driving the optical path adjustment element to move relative to the fixed portion; wherein when viewed along a direction that is parallel with the main axis, the anti-vibration driving mechanism is disposed on the second side, and the anti-vibration driving mechanism and the driving mechanism at least partially overlap; wherein the anti-vibration driving mechanism drives the optical path adjustment element to rotate in a first rotating axis relative to the fixed portion, and the first rotating axis is parallel with the second side.

18. An optical system, comprising: an optical module with a main axis, comprising: a fixed portion; a movable portion connected to an optical element and movable relative to the fixed portion; a driving mechanism driving the movable portion to move relative to the fixed portion; a supporting assembly connected to the movable portion and the fixed portion; and a heat dissipation assembly configured to improve the heat dissipation efficiency of the optical element and the driving mechanism; wherein when viewed along a direction that is parallel with the main axis, the fixed portion is a polygonal structure with a first side, a second side, a third side, and a fourth side; wherein the first side is parallel with the third side, the second side is parallel with the fourth side, and the first side is not parallel with the second side; wherein the fixed portion comprises an outer frame and a base having a plate-like structure, the base is perpendicular to the main axis, the base and the outer frame are arranged along the main axis, and the outer frame comprises: a top wall having a plate-like structure, parallel with the base, and not parallel with the main axis, wherein the top wall is parallel with the base; a first side wall extending from a first edge of the top wall and not parallel with the top wall, wherein when viewed along a direction that is parallel with the main axis, the first side wall is disposed on the first side; a second side wall extending from a second edge of the top wall and not parallel with the top wall, wherein when viewed along a direction that is parallel with the main axis, the second side wall is disposed on the second side; a third side wall extending from a third edge of the top wall and not parallel with the top wall, wherein when viewed along a direction that is parallel with the main axis, the third side wall is disposed at the third side; and a fourth side wall extending from a fourth edge of the top wall and not parallel with the top wall, wherein the fourth side wall has an opening corresponding to a light, the opening is formed between the outer frame and the base, and wherein when viewed along a direction that is parallel with the main axis, the fourth side wall is disposed on the fourth side; wherein the movable portion comprises: a pedestal connected to the optical element and having a plate-like structure; and a holder fixedly disposed on the pedestal; wherein the heat dissipation assembly comprises: a first heat dissipation element having a plate-shaped structure, having a material of carbon fiber, ceramic or metal material, and having a thermal conductivity greater than $10 \text{ W}/(\text{m}\times\text{K})$, wherein the first heat dissipation element and the base have an integral structure, and the first heat dissipation element is closer to the optical element than the top wall; and a second heat dissipation element having a plate-shaped structure, having a magnetically permeable material to improve the driving efficiency of the driving mechanism, and having a thermal conductivity greater than $10 \text{ W}/(\text{m}\times\text{K})$, wherein the second heat dissipation element is fixedly disposed on the pedestal and at least partially embedded in the pedestal, and the second heat dissipation element corresponds to the driving mechanism to generate an attractive force or a repulsive force; wherein the second heat dissipation element is parallel with the first heat dissipation element; wherein a gap greater than zero is formed between the first heat dissipation element and the second heat dissipation element.

19. The optical system as claimed in claim 18, wherein the optical module further comprises a frame connected to the outer frame, and a first accommodating space is formed to accommodate the movable portion and the frame.

20. The optical system as claimed in claim 19, wherein the optical module further comprises an optical path adjustment element connected to the frame, and a second accommodating space is

formed between the frame and the movable portion to accommodate the optical path adjustment element.
